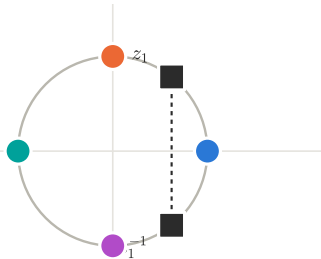


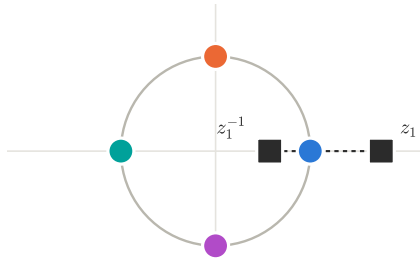
(a) $t = 4$, one pair, $|z| = 1$



on the circle, and free of the orbit

$$\prod(\text{alphabet}) = (-1)^{t-1} = -1$$

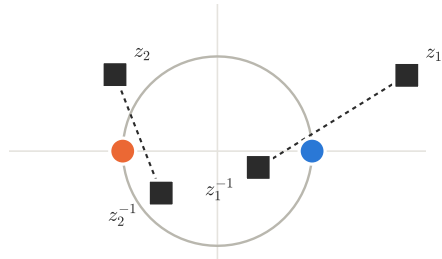
(b) $t = 4$, one pair, z real



z outside, z^{-1} inside: a free direction

$$\prod(\text{alphabet}) = (-1)^{t-1} = -1$$

(c) $t = 2$, two pairs



an r -dimensional free torus, inside $O(6)^-$

$$\prod(\text{alphabet}) = (-1)^{t-1} = -1$$

● the frozen orbit μ_t ■ a free reciprocal pair z, z^{-1}